Science fiction tends to confuse intelligence with consciousness, and assume that in order to match or surpass human intelligence, computers will have to develop consciousness. The basic plot of almost all movies and novels about AI revolves around the magical moment when a computer or a robot gains consciousness. Once that happens, either the human hero falls in love with the robot, or the robot tries to kill all the humans, or both things happen simultaneously.

But in reality, there is no reason to assume that artificial intelligence will gain consciousness, because intelligence and consciousness are very different things. Intelligence is the ability to solve problems. Consciousness is the ability to feel things such as pain, joy, love and anger. We tend to confuse the two because in humans and other mammals intelligence goes hand in hand with consciousness. Mammals solve most problems by feeling things. Computers, however, solve problems in a very different way.

There are simply several different paths leading to high intelligence, and only some of these paths involve gaining consciousness. Just as airplanes fly faster than birds without ever developing feathers, so computers may come to solve problems much better than mammals without ever developing feelings. True, AI will have to analyse human feelings accurately in order to treat human illnesses, identify human terrorists, recommend human mates and navigate a street full of human pedestrians. But it could do so without having any feelings of its own. An algorithm does not need to feel joy, anger or fear in order to recognise the different biochemical patterns of joyful, angry or frightened apes.

Of course, it is not absolutely impossible that AI will develop feelings of its own. We still don't know enough about consciousness to be sure. In general, there are three possibilities we need to consider:

1. Consciousness is somehow linked to organic biochemistry in such a way that it will never be possible to create consciousness in non-organic systems.
2. Consciousness is not linked to organic biochemistry, but it is linked to intelligence in such a way that computers could develop consciousness, and computers will *have to* develop consciousness if they are to pass a certain threshold of intelligence.
3. There are no essential links between consciousness and either organic biochemistry or high intelligence. Hence computers might develop consciousness – but not necessarily. They could become super-intelligent while still having zero consciousness.

At our present state of knowledge, we cannot rule out any of these options. Yet precisely because we know so little about consciousness, it seems